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# FIND-S Algorithm Implementation

def find_s(training_data):
    """
    training_data: list of tuples (attributes, label)
    attributes: list of attribute values
    label: 'Yes' (positive) or 'No' (negative)
    """
    # Step 1: Initialize hypothesis with the first positive example
    for attributes, label in training_data:
        if label == "Yes":
            hypothesis = attributes[:]
            break

    # Step 2: For each positive example, update hypothesis
    for attributes, label in training_data:
        if label == "Yes":
            for i in range(len(hypothesis)):
                if hypothesis[i] != attributes[i]:
                    hypothesis[i] = "?"

    return hypothesis

# Example Training Data
# Attributes: [Sky, AirTemp, Humidity, Wind, Water, Forecast]
training_data = [
    ("Sunny", "Warm", "Normal", "Strong", "Warm", "Same", "Yes"),
    ("Sunny", "Warm", "High", "Strong", "Warm", "Same", "Yes"),
    ("Rainy", "Cold", "High", "Strong", "Warm", "Change", "No"),
    ("Sunny", "Warm", "High", "Strong", "Cool", "Change", "Yes")
]

# Run FIND-S
final_hypothesis = find_s(training_data)

print("Final Hypothesis:", final_hypothesis)

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Final Hypothesis: ['Sunny', 'Warm', '?', 'Strong', '?', '?']
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